Open-chord transfer and spectral towers

The same-disorder Slater representation in complex double-scaled SYK

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Abstract

We turn the established joint moment law into an explicit transfer on ordered open endpoints. The Hamiltonian has a q-oscillator creation part; its orthogonal primitive subspaces generate standard q-Jacobi towers. The selected creation block is compact in the full joint representation, reconstructs the grading through its deformed commutator, and admits canonical undressing on the graded core. The joint weak closure is Type I. The spectral measure of a single selected excitation is exactly $(1-\sqrt{q}+\sqrt{q}E^2)\,d\mu_q(E)$, giving finite-time returns and Euclidean preparation kernels. A separate two-energy kernel describes compression to the vacuum Hamiltonian tower. These are explicit model results; physical observer-state and bulk-generator identifications remain to be made.

1 Microscopic input and existing constructions

The preceding stage proved the all-fixed-word law for the same Hermitian Gaussian p-body Hamiltonian H_N and its Slater-adapted block $M_N^\dagger$. Explicitly, with standard complex CAR,

$$H_N = \sqrt{w_N} \sum_{|A|=|B|=p} J_{AB} C_A^\dagger C_B, \qquad w_N = \frac{4^p J_s^2}{\lambda_N} \left(\frac{p!}{N^p}\right)^2, \tag{1}$$

$$h_N = \frac{H_N}{\sqrt{w_N D_N}}, \qquad M_N^\dagger = \frac{1}{\sqrt{L_N}} \sum_{A\subset E,\ B\subset O} J_{AB} C_A^\dagger C_B. \tag{2}$$

Here $m = N/2$, $|O| = |E| = m$, all subsets have size p, $L_N = \binom{m}{p}^2$, $D_N = \binom{m}{p}\binom{m+p}{p}$, and $\mathbb{E}J_{AB}J_{CD} = \delta_{AD}\delta_{BC}$. Overlapping sets are included. The state is $\varphi_N(X) = \mathbb{E}\langle O|X|O\rangle$. Take $p \to \infty$, $\lambda_N = p^2/N \to \lambda > 0$, fixed $J_s > 0$, $\hbar = 1$, and string units for H_N. Set $q = e^{-4\lambda}$, $s = \sqrt{q}$, $\gamma = q^{1/4}$. All transfer energies below are for dimensionless h; physical string energies multiply by $J_s/\sqrt{\lambda}$ in this limit.

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The input word law assigns γ to each mixed h–M edge, q to every crossing, and q to each nesting whose inner edge is selected. Only compatible forward selected edges survive. The preceding proof also established actual finite-N annealed finite-time convergence on compact time sets.

Open-chord transfer, q-Wick recursion and dressed oscillators are established constructions [1, 2, 3]. In particular, Rajgadia–Xu map the Hamiltonian to a chord creation plus annihilation operator [2, Sec. 3.2]. The task here is to implement the same-J mixed contractions: the off-diagonal source $\gamma q^{\mathcal{N}}$ below is retained. We do not claim the general transfer or Type I strategy as a new idea.

2 Transfer on ordered open endpoints

A formal basis vector $|w\rangle$ is a string $w = t_1 \cdots t_n$, ordered from oldest to newest open endpoint. Letters t_j are H or P, meaning an opening by h or $M^\dagger$. Write $w \setminus j$ for deletion of position j. Define creation maps

$$a^\dagger|w\rangle = |wH\rangle, \qquad b^\dagger|w\rangle = |wP\rangle, \tag{3}$$

and annihilation maps, zero on the empty string,

$$a|w\rangle = \sum_{j:t_j=H} q^{n-j}|w \setminus j\rangle + \gamma q^{n-1} \sum_{j:t_j=P} |w \setminus j\rangle, \tag{4}$$

$$b|w\rangle = q^{n-1}\left(\gamma \sum_{j:t_j=H} |w \setminus j\rangle + \sum_{j:t_j=P} |w \setminus j\rangle\right). \tag{5}$$

The physical operators in the limiting representation are

$$h = a + a^\dagger, \qquad M = b, \qquad M^\dagger = b^\dagger. \tag{6}$$

Evaluate a word by applying these maps from right to left to $|\varnothing\rangle$ and extracting its empty-string coefficient.

To prove the transfer rule, close an edge at position j while n edges remain open. The $n - j$ newer edges cross it; the $j - 1$ older ones enclose it. A general closing edge therefore receives q^{n-j}, while a selected closing edge receives q^{n-1}. A closing M always selects its edge. A closing h selects its edge only when the open endpoint was P. Mixed endpoints supply γ. Every pair of overlapping edges is counted exactly when the first of them closes, so (4) reproduces every diagram and hence all joint moments. The status of an edge need not be guessed when it opens.

3 The positive metric, rather than a Euclidean string norm

Different string lengths are orthogonal. For equal-length strings u, v, define $G_n(u, v)$ by closing the letters of u in reverse order against v, using a for H and b for P, and taking the empty coefficient. Equivalently, sum over all complete pairings between the two boundaries with the established joint weights. Reflection of such a diagram preserves its weight, so G_n is Hermitian. Moving the last endpoint across the cut shows

$$\langle u, a^\dagger v\rangle = \langle au, v\rangle, \qquad \langle u, b^\dagger v\rangle = \langle bu, v\rangle. \tag{7}$$

This metric is generally not the identity matrix on strings.

For completeness, positivity does not require guessing a Fock metric. Recursively define polynomials χ_w in $h, M^\dagger$:

$$\chi_\varnothing = I, \qquad \chi_{wP} = M^\dagger \chi_w, \tag{8}$$

$$\chi_{wH} = h\chi_w - \sum_v a_{v,w}\chi_v, \qquad a|w\rangle = \sum_v a_{v,w}|v\rangle. \tag{9}$$

The transfer sends $\chi_w\Omega$ exactly to $|w\rangle$. Using (7) and vacuum word matching gives

$$G(u, v) = \varphi(\chi_u^* \chi_v). \tag{10}$$

The right side is a limit of positive microscopic Gram matrices. Thus $G \geq 0$; quotient its null space and complete. The maps preserve null vectors by adjunction and descend to the previously constructed GNS representation. Each level $\mathcal{H}_n$ has dimension at most 2^n. No assertion that all formal strings are independent is needed. Let $\mathcal{N}$ be the resulting grading operator.

Cost with finitely many selected insertions

During propagation by h, an H can be appended or any endpoint can close; no new P is appended. With at most r selected creation insertions, level n therefore uses at most

$$\sum_{j=0}^{\min(r,n)} \binom{n}{j} \tag{11}$$

formal strings. This is polynomial in n at fixed r. Each state has at most $n + 1$ transitions. Repeated endpoints leading to the same string must have their coefficients added. This gives a sparse, direct evaluation algorithm for fixed-insertion time series, with the preceding stage's uniform tail bound.

4 Exact algebra, bounded operators, and canonical undressing

Appending a letter and then applying (4) gives

$$aa^\dagger - qa^\dagger a = I, \qquad ab^\dagger - qb^\dagger a = \gamma q^{\mathcal{N}}, \tag{12}$$

$$ba^\dagger - qa^\dagger b = \gamma q^{\mathcal{N}}, \qquad bb^\dagger - qb^\dagger b = q^{\mathcal{N}}. \tag{13}$$

Both creators raise $\mathcal{N}$ by one. These relations are on the full graded core, not just the pure-M ladder. By induction in the level and adjunction,

$$\|a^\dagger|_{\mathcal{H}_n}\|^2 \le [n+1]_q, \qquad \|b^\dagger|_{\mathcal{H}_n}\|^2 \le (n+1)q^n, \qquad [n]_q = \frac{1-q^n}{1-q}. \tag{14}$$

For example the second estimate follows from $\|b^\dagger v\|^2 = q\|bv\|^2 + q^n\|v\|^2$. The pure H and pure P strings attain the respective bounds. Consequently a, b are bounded, and

$$\|a\| = (1-q)^{-1/2}, \qquad \|M\|^2 = \max_{n\ge 0}(n+1)q^n. \tag{15}$$

Moreover b is compact: its level norms tend to zero and each level is finite dimensional. The grading is reconstructible from the physical selected block:

$$\mathcal{Q} = MM^\dagger - qM^\dagger M = q^{\mathcal{N}}. \tag{16}$$

Define on the finite-level core

$$\beta = q^{-\mathcal{N}/2}b, \qquad \beta^\dagger = b^\dagger q^{-\mathcal{N}/2}. \tag{17}$$

Then

$$[\beta, \beta^\dagger] = I, \qquad a\beta^\dagger - \sqrt{q}\,\beta^\dagger a = \gamma q^{\mathcal{N}/2}. \tag{18}$$

The inverse dressing is unbounded. This is an exact statement in the limiting graded representation, not an operator-norm identity for finite microscopic matrices or an unproved identification of $\mathcal{N}$ with microscopic occupation in the full joint sector.

Since $\mathcal{Q}$ has simple top eigenvalue one at Ω, $\mathcal{Q}^r \to |\Omega\rangle\langle\Omega|$ in operator norm. The bounded joint algebra contains that rank-one projection. Cyclicity then produces all rank-one operators between polynomial vectors. Therefore

$$W^*(h, M) = B(\mathcal{H}). \tag{19}$$

Its vacuum vector state is pure. This implements the known state-adapted Type I mechanism [2] in the same-J joint model.

5 Orthogonal Hamiltonian towers

Let $\mathcal{K}_d = \ker(a|_{\mathcal{H}_d})$ and $\mathcal{K} = \bigoplus_{d\geq 0} \mathcal{K}_d$. The first relation in (12) implies that $a^\dagger$ is injective and has closed range at each level. Hence

$$\mathcal{H}_n = a^\dagger \mathcal{H}_{n-1} \ \perp \ \mathcal{K}_n. \tag{20}$$

For a unit vector $\xi \in \mathcal{K}$, its tower has normalized vectors

$$|r;\xi\rangle = \frac{(a^\dagger)^r \xi}{\sqrt{[r]_q!}}, \qquad a(a^\dagger)^r \xi = [r]_q (a^\dagger)^{r-1} \xi. \tag{21}$$

Orthogonal primitive vectors generate orthogonal towers, including primitives of different original grades. Level induction spans the whole space. Thus

$$\mathcal{H} \simeq \ell^2(\mathbb{N}_0) \otimes \mathcal{K}, \qquad h \simeq J_q \otimes I_{\mathcal{K}}, \tag{22}$$

where

$$J_q|r\rangle = \sqrt{[r+1]_q}|r+1\rangle + \sqrt{[r]_q}|r-1\rangle. \tag{23}$$

No formula for the dimensions of $\mathcal{K}_d$ is assumed. The full joint dynamics remains coupled through M, which moves between these towers.

A constructive finite-level projection onto $\ker a$ is

$$\Pi|_{\mathcal{H}_n} = \sum_{j=0}^{n} \frac{(-1)^j q^{j(j-1)/2}}{[j]_q!} (a^\dagger)^j a^j. \tag{24}$$

On each tower this is the finite q-binomial identity, equal to one at height zero and zero otherwise. Every $v \in \mathcal{H}_n$ has the explicit decomposition

$$v = \sum_{r=0}^{n} \frac{(a^\dagger)^r}{[r]_q!} \Pi a^r v. \tag{25}$$

This supplies both a theoretical decomposition and an exact algorithm; it was checked coefficient by coefficient on all formal strings through level four at two rational parameters.

The measure of J_q at height zero is the standard normalized q-Gaussian μ_q. Its support is $[-2/\sqrt{1-q}, 2/\sqrt{1-q}]$, and $\|h\| = 2/\sqrt{1-q}$. Writing $E = 2\cos\theta/\sqrt{1-q}$,

$$d\mu_q = \frac{(q;q)_\infty |(e^{2i\theta};q)_\infty|^2}{2\pi}\, d\theta, \qquad 0 \leq \theta \leq \pi. \tag{26}$$

This is the established continuous q-Hermite spectral measure [3]; the new application is the primitive decomposition of the same-J selected sector.

6 An exact selected-excitation spectral measure

The selected one-excitation vector has the orthogonal split

$$b^\dagger\Omega = \gamma a^\dagger\Omega + \eta, \qquad \eta = (b^\dagger - \gamma a^\dagger)\Omega, \qquad a\eta = 0, \quad \|\eta\|^2 = 1 - s. \tag{27}$$

Thus η starts a second standard q tower, orthogonal to the entire vacuum Hamiltonian tower. Directly in string coordinates,

$$z_n = |PH^n\rangle - \gamma|H^{n+1}\rangle, \qquad hz_n = z_{n+1} + [n]_q z_{n-1}, \quad z_0 = \eta. \tag{28}$$

For every bounded Borel function f,

$$\boxed{\varphi(Mf(h)M^\dagger) = \int (1 - s + sE^2) f(E)\, d\mu_q(E).} \tag{29}$$

The measure is positive and has unit mass because $\int E^2 d\mu_q = 1$. It is supported on the same energy interval as the reference state but redistributes its spectral weight.

Let $R(t) = \int e^{itE} d\mu_q(E)$. Then

$$C_M(t) = \varphi(Me^{ith}M^\dagger) = (1 - s)R(t) - sR''(t), \qquad \varphi(e^{ith}M^\dagger) = -i\gamma R'(t). \tag{30}$$

The second identity retains the same-disorder cross amplitude. Equation (29) independently recovers the previous second and fourth moments with two selected endpoints.

For an explicit preparation, define

$$|\psi_\beta\rangle = Z_M(\beta)^{-1/2} e^{-\beta h/2} M^\dagger\Omega, \qquad Z_M(\beta) = \int (1 - s + sE^2) e^{-\beta E} d\mu_q(E). \tag{31}$$

If $Z(\beta) = \int e^{-\beta E} d\mu_q(E)$, then

$$Z_M = (1 - s)Z + sZ'', \qquad \langle h\rangle_{\psi_\beta} = -\partial_\beta \log Z_M, \qquad \langle e^{ith}\rangle_{\psi_\beta} = \frac{Z_M(\beta - it)}{Z_M(\beta)}. \tag{32}$$

These are normalized vector-preparation formulas. They do not identify a Gibbs/KMS observer state on the full joint algebra. The spectral theorem establishes them in the limiting model; the preceding finite-time proof transfers the exponential matrix coefficients from the actual finite-N annealed theory.

7 A two-energy kernel on the vacuum tower

Let P_0 project onto the closed h-cyclic subspace of Ω, not onto the one-dimensional vacuum. Its orthonormal basis is $|n\rangle = (a^\dagger)^n\Omega/\sqrt{[n]_q!}$. Counting the one selected edge among all cross-boundary pairings gives

$$\langle H^{n+1}, H^n P\rangle = \gamma(n+1)q^n[n]_q!. \tag{33}$$

There are $n+1$ choices of its partner, all with n overlap penalties; the remaining general-edge matchings give $[n]_q!$. Consequently

$$P_0M^\dagger P_0|n\rangle = d_n|n+1\rangle, \qquad d_n = \frac{\gamma(n+1)q^n}{\sqrt{[n+1]_q}}. \tag{34}$$

Let $p_n(E)$ be the orthonormal polynomials of μ_q. The two-energy integral kernel is

$$B(E,E') = \sum_{n\geq 0} d_n p_{n+1}(E)p_n(E'), \qquad B \in L^2(\mu_q \otimes \mu_q). \tag{35}$$

For example this computes $\varphi(e^{ith}M^\dagger e^{iuh})$ by integrating $e^{itE}B(E,E')e^{iuE'}$ against the two measures. It also computes a single selected insertion between two Euclidean-prepared reference vectors.

The Hilbert–Schmidt tail after $n = L-1$ obeys, with $r = q^2$,

$$\sum_{n\geq L}|d_n|^2 \leq sr^L\left[\frac{(L+1)^2}{1-r} + \frac{2(L+1)r}{(1-r)^2} + \frac{r(1+r)}{(1-r)^3}\right]. \tag{36}$$

This follows from $[n+1]_q \geq 1$ and the exact geometric sum of $(n+1)^2r^n$; it controls the spectral tail without neglecting cancellations in a time integral.

Repeated insertions require the full transfer. In particular, $\|P_0M^\dagger\Omega\|^2 = s$ while $\|M^\dagger\Omega\|^2 = 1$. Iterating only (35) would discard the primitive component η and yield an incorrect four-point function. The one-insertion kernel and the full joint transfer have distinct, explicit roles.

8 Numerical spectra and preparation examples

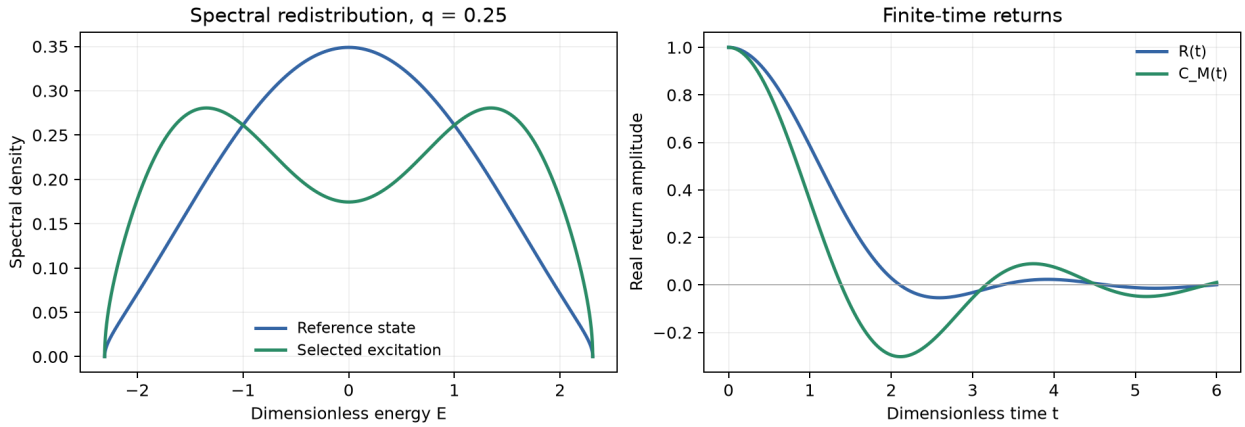


Figure 1: Dimensionless $q = 0.25$ spectral densities and return amplitudes from (29)–(30). The selected excitation redistributes weight toward larger $|E|$. The plotted times are fixed times in the limiting annealed representation.

The numerical calculation diagonalizes the real symmetric q-Jacobi matrix at sizes 160 and 240, using its vacuum spectral weights. It independently evaluates (26) with 256 product factors and 8,193 angular samples; normalization and variance agree with one within 10^{-10}. Fifteen selected returns at three q values agree between the two Jacobi sizes to about 1.4×10^{-15} or better.

For illustration, the filtered selected vector at $\beta = 2$ has the following energies in normalized h units:

q	$Z_M(2)$	$\langle h \rangle$	$\mathrm{Var}(h)$
0.0625	6.3522341	-1.4597969	0.2546408
0.25	8.9789308	-1.6651661	0.2437047
0.64	17.3664951	-2.1157408	0.3337576

These values describe Euclidean filtering, not a temperature assigned to the full nonstationary state.

For an L-node Gaussian quadrature, polynomial exactness and the compact support radius $E_* = 2/\sqrt{1-q}$ give the selected-return error bound in exact arithmetic

$$2e^{|t|E_*} \frac{(|t|E_*)^{2L-2}}{(2L-2)!}. \tag{37}$$

The degree-two selected weight explains the $2L-2$ exponent. The saved numerical record separates this analytic truncation bound from observed floating-point differences and imaginary roundoff. It is not an interval certificate for the eigensolver. At $t = 0$ the saved logarithmic bound is $-\infty$, denoting an exactly zero Taylor truncation bound.

9 Verification, preserved failure, and the next physics task

The new transfer checker uses exact rational $\gamma = 1/2, 2/3$. It compares 1,458 moments with the sealed sixth-word polynomials and 232 eighth-word cases with independent pairing enumeration. It additionally checks 248 algebra relations, 124 primitive projection identities, 62 full tower decompositions, 680 adjunction entries, 170 metric symmetries and 54 residual-tower, compression and selected-spectral identities. A separate checker verifies 62 grading identities, 62 canonical commutators, 62 same-J cross relations and 18 sharp selected-shift norms.

The initial checker failed before producing a scientific output: Python integer division at the zero-level factorial introduced a floating coefficient and a residual of about 8.7×10^{-19}. The failed source and diagnosis are preserved. Changing that coefficient to exact rational division resolved the failure; the analytic decomposition was unchanged. Successful scientific outputs were then sealed and not rerun. The three output records, figure, parameters and environment versions are under `results/susskind-slater-transfer/`; the evidence manifest binds these to the document, two-pass build and page renders.

The transfer now supplies all joint correlators without explicit pairing enumeration, while the primitive decomposition gives a transparent Hamiltonian spectrum and selected preparation kernel. The next target is to identify physically meaningful states and observables in this explicit representation and compare their propagation with the static-patch bulk requirements. In particular, the grading reconstructed from M needs a microscopic joint-sector interpretation before it is called a spatial or radial observable. Mass, locality, stationary observer conditions and finite-G interactions remain tests to perform. Type I closure alone does not establish that dictionary or a finite horizon entropy.

References

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